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An Adaptive and Integrated Prediction of Taxi Trajectories using Regression Model

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***Abstract*—Taxi trajectory prediction can help many advanced location-based services, such as advertisements for passengers. Previous approaches, which consider trajectories as one-dimensional patterns and process them in single scales, does not consider two-dimensional patterns of trajectories in various spatial scales. In road network analysis, one of the most main problems is finding important crossroads, which can be useful in transportation planning. However, none of the previous approaches identifies the problem of finding network-wide important crossroads in the actual road network. This paper proposes a strategic decision based on data analysis and interpretation, based approach to find important crossroads. This method is to model the road trip network focusing on real time travel demands using a tripartite graph, instead of completely analyzing on the topology of road network. A novel ranking algorithm that propagates scores along a tripartite graph is proposed to find the importance scores of crossroads accurately.**

Keywords– Origin Destination (OD), Taxi trajectories, Random Forest Regressor, Gradient Boosting Regressor, Linear Regression, K Neighbors Regressor, Decision Tree Regressor, Multi Output Regressor, Grid search cv.

I. INTRODUCTION

Transportation by taxis have become an important part of many cities around the world for residents going to work, shopping, and other travel activities. As well as the difficulty of parking associated with self-driving, it is often more comfortable and easier to use than other travel options. The traditional taxi sector has been dramatically changed by the advent of the Internet and mobile applications, as well as developing reservation systems and ride-hailing automobile services. In [1], they use previous trajectory data, to determine which road segments are reachable from a given location at a given time. The Intersection-of-Multi-location Query Maximum Bounding region search (I-MQMB) algorithm and the I-Query Trace Back Search (I-TBS) algorithm are proposed to effectively answer I-Queries. Choosing locations that covers the maximum number of trajectories around the spatial region is not focused on this paper [2]. On the other hand, due to increased demand for mobility, finding available taxis throughout long periods of time continues to be a challenge. Specifically, finding a vacant cab in a lot of commercial locations, which may include huge office and shopping districts, can be extremely challenging during peak periods [3]. Clearly, merely increasing the number of cabs on the road does not appear to be an effective solution to the persistent demand-supply gap in taxi services. Furthermore, it not only worsens urban congestion but also has an impact on environmental quality. By using massive taxi trajectories as a feature of a crossroads ranking, we present efficient method that differ from traditional approaches. This method uses the dynamic information of the road network [4]. The outcome shows that, when compared with the baseline algorithm, this system is efficient to find the main crossroads with high connectivity, especially, the crossroads which are located on the ring roads, where a wider range of Origin-Destination pairs begins and ends, thus obtain more importance score. This can achieve a well-balanced tradeoff among various parameters [5]. Environment problems are solved on an end-to-end basis and achieve sub-optimal performance. It is a fast and easy procedure to perform. As part of developing route network, we combine route knowledge of drivers, travel demands, as well as traffic information. A tripartite graph is constructed based on information from the information extraction phase. An algorithm is proposed to rank the important crossroads by using score propagation over the tripartite graph [6]. This method achieves a well-balanced tradeoff among various parameters. The algorithms used are: Random Forest Regressor and Lasso Algorithm. These algorithms Improves Interpretability, Regularization Improves Accuracy and Generalization and has Automatic Variable Selection. This method has the best acceptance ratio and it’s very quick and efficient to use. It performs better on various circumstances and environment. The problems are solved on an end-to-end basis. It achieves sub-optimal performance. It is a fast and easy procedure to perform [7].

II. RELATED WORK

A large number of historical trajectories is required to create a prediction model, because some rural areas lack coverage of these services due to lack of privacy concerns [8]. It requires more urban applications to refine the attainable area's boundaries [9].Data changes may cause the optimal model to change significantly, even a small change in the data can have a dramatic effect [10]. The results demonstrate not only the proposed approach's efficiency and scalability, but also the financial benefits it provides. However, the real time dataset reveals that a high level of uncertainty is present in sensor values. The model's efficiency does not increase as the number of vehicles increases, hence it is sensitive to the number of vehicles. The main benefit of this model which is trained offline is that it is quick to process queries based on traffic speed distribution. However, this method usually necessarily involves the use of a Differential Global Positioning System (DGPS) in order to minimize the GPS errors in data collection. However, due to its expensive price, DGPS is not extensively used in consumer-grade mobile devices. In Real-time forecasting is achievable according to the model's performance. When compared to baselines that use historical trajectories, the proposed model offers significantly better prediction accuracy and timeliness in projecting statistically significant gathering occurrences. The spatial co-location mining and MapReduce paradigms are discussed in this study, as well as the search space partition technique and Co-location mining algorithm based on MapReduce. Parallelism is beneficial since it aids in dividing the search space for co-location patterns into distinct portioned data. In contrast to the prior state-of-the-art technology, it provides highly quick query execution, both random and positive searches show significant gains in processing time. The resultant index can readily expand to massive-size graphs thanks to a large number of tests, even when several state-of-the-art indexes fail to finish the construction. A new approach is described for identifying a chain decomposition of a Directed Acyclic0 Graph, which can be used to reduce transitive closures. It outperforms all other graph labelling methods in terms of efficiency. This paper discusses the confidentiality location-based database server, In this way, location based database server capabilities can be offered at a lower quality, while protecting sensitive private information of mobile users. When the Location Anonymizer receives precise location information from a mobile user, it converts it into a masked spatial regions. It's possible that the location-based database server won't be able to give its users with a high-quality service.

III. METHODOLOGY

Using a data-driven framework, we aim to identify key nodes in the road network. In the existing work, process of mining spatio-temporal accessible areas aims to identify which roads are approachable from the given places within a certain temporal period based on previous trajectories, by using a collection of user-specified locations and a big trajectory dataset shown in Figs 1 and 2. The recent work developed efficient methods for the Single-location spatio-temporal reachability Query (S-Query) and the Union-of-multi-location spatio-temporal reachability Query (U-Query). They analyzed a large dataset of real taxi trajectory routes in Shenzhen, China. These results demonstrate that the proposed approach, when compared with the baseline method, significantly reduces I-Queries running time by 50%.

The direction-based trajectory representation method is used in the proposed model, which replaces both the input and output sequences of road segments with sequences of intersections and related movement directions. For autonomous driving applications, a number of related techniques are being developed. They normally concentrate on a single section of road and forecast the situation of surrounding vehicles in the next seconds. Due to the vastly different time and spatial scales, these models are inapplicable to the challenge of predicting vehicle trajectory at the city scale. Only a few recent studies have attempted to forecast a vehicle's path choice as it approaches an intersection by specifying a set of intentions such as travelling straight, turning left, turning right, and so on. Instead of just examining the architecture of the road network, we use a tripartite graph to construct a realistic depiction for road trips that focuses on real travel demands. A novel ranking algorithm that propagates scores along a tripartite graph is proposed to calculate the junction of road's accuracies importance score. We suggest a trajectory modeling method based on direction, in which each road segment is represented as a pair of crossings is represented as a pair of intersections with corresponding movement directions. This reduces the dimensionality of the forecast from over 1,000,000 road segments to a small number of direction identifiers, improving modelling efficiency and prediction accuracy. Instead of using a static road network, we use a trip network that includes the flow of traffic, source-destination, and route paths are all elements of dynamic transportation data. We first gather trip data from enormous taxi trajectories and the road network, and then use a tripartite graph to describe the trip network. A tripartite graph is one in which the nodes may be divided into three distinct sets, with no two nodes in the same set being connected directly. A pair of adjacency matrices can be used to describe the connections between two sets of nodes as encoded by their edges. Using data from the information extraction phase, we create a tripartite graph to model a travel network. We must find the relationship between transitions, pathways, and crossroads in order to construct the tripartite graph. A transition can be made based on the regions where a trajectory's starting and destination roads are located. The different advantages of using this method are, it obtains a well-balanced tradeoff between different parameters. It has the highest rate of acceptance. It's efficient to use and quick to implement. This achieves sub-optimal results. It performs a simple procedure to carry out the operation. It works better on various circumstances and environment. The difficult tasks are solved on an end-to-end basis. We undertake data preprocessing to increase calculation accuracy and efficiency.

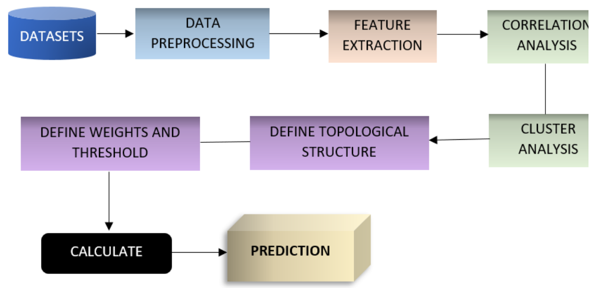

Fig 1. An overview of framework

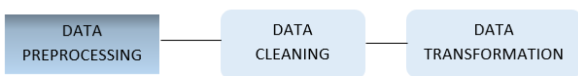

Fig 2. Process of data pre-processing where the dataset is cleaned and transformed from unstructured, noisy and irrelevant data.

Dataset:
We're using dataset collected from Kaggle. This has nearly 1.7M observations of taxi trips.

Module 1: Data Pre-processing
The dataset is collected from Kaggle. Data preprocessing is done where data cleaning approach is performed to remove all the unwanted data from the dataset. Data Transformation makes the collected data into a unique format that's appropriate for mining technique. The dataset is usually unstructured, noisy, and irrelevant. So this data must go through a data preprocessing phase before it can be used for pattern mining and pattern analysis.

Module 2: Exploratory data analysis
By preprocessing data, we improve it's quality while also reducing its size, Out Exploratory data analysis revealed that it contained missing GPS updates and incorrect information. In fact, it's missing value column could be viewed as one of the dataset's short coming. We identified missing GPS points in the trajectory by scanning for huge gaps between two consecutive GPS updates, i.e., distances greater than a cab would have travelled if there had been a speed limit. The information about trip start timestamps was also inconsistent; for example, several of the trips in the test set have starting timestamps that are hours before the cutoff timestamps of the relevant snapshot with only a few GPS updates. Moreover, some taxi trips have many uncommon trajectories, e.g., Some taxi drivers neglected to turn off the metre after completing a ride, such as when returning to the central city area after dropping a passenger at the airport; such turning back routes were exceedingly difficult to foresee.

Module 3: Time-series visualization
We'll visualize trips taken for each month in a year to analyze which month has the highest number of trips and lowest number of trips, Trips taken on days for each month to find which time of the month is at peak and which dates are at the lowest, Trips taken on weekdays and which call type has taken the most number of trips to find which days are the busiest.

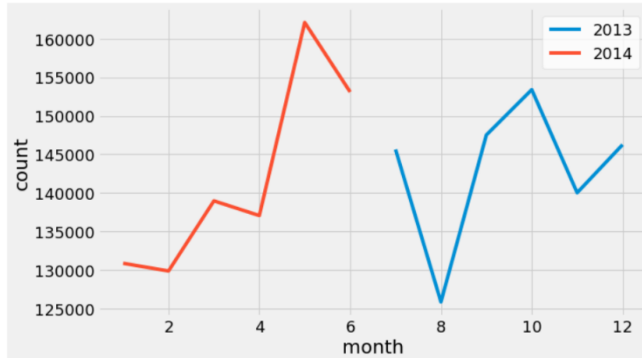

Fig 3. Visualization per Month count.

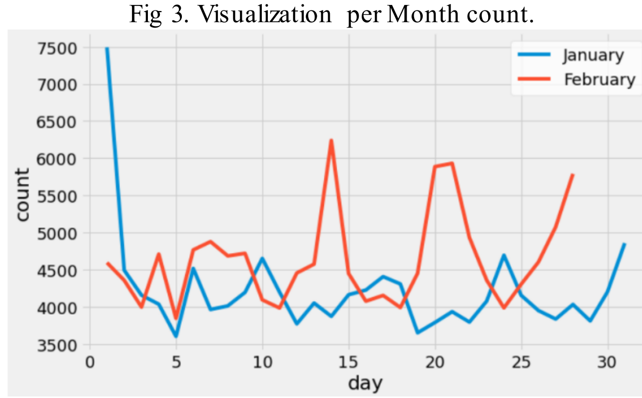

Fig 4. Visualization of trips for January and February.

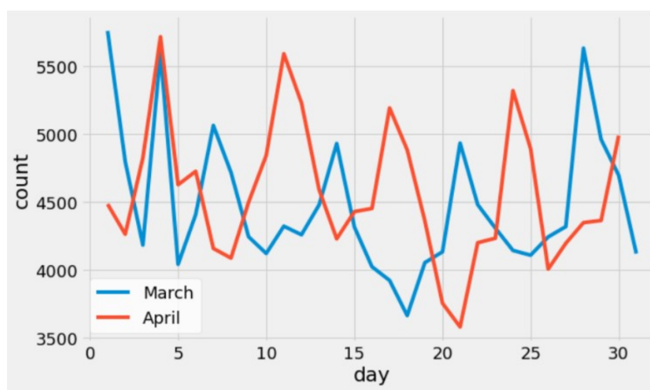

Fig 5. Visualization of trips for March and April.

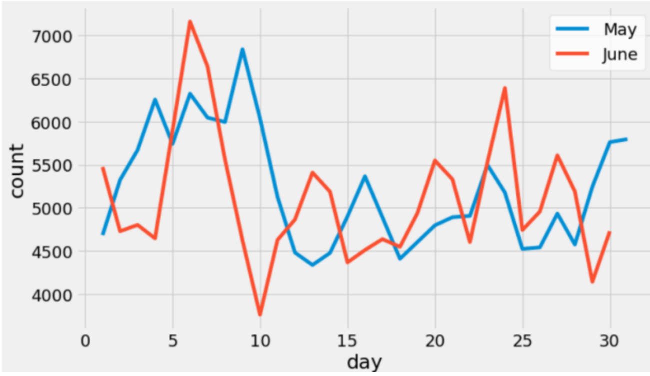

Fig 6. Visualization of Trips for May and June.

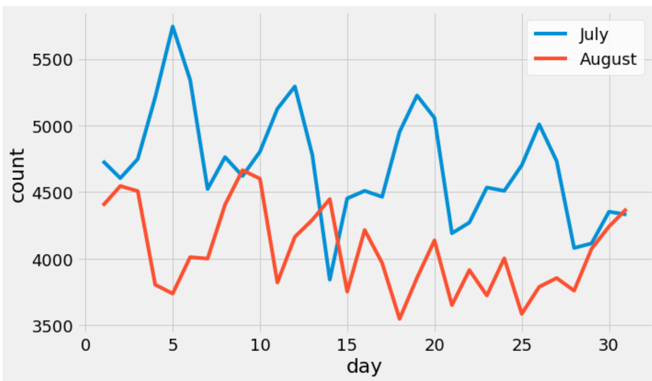

Fig 7. Visualization of Trips for July and August.

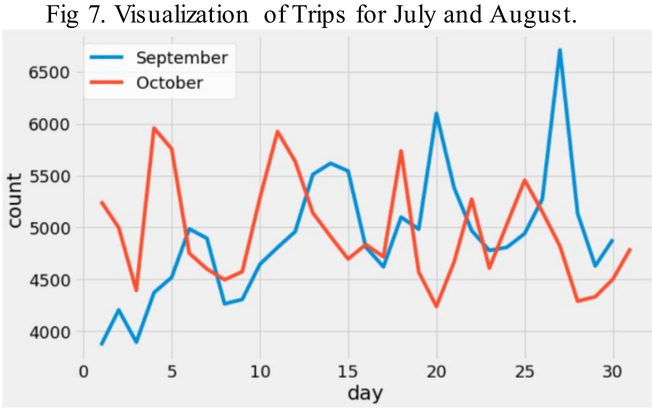

Fig 8. Visualization of Trips for September and October.

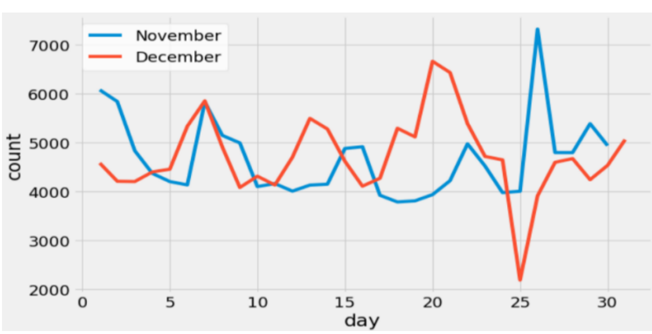

Fig 9. Visualization of Trips for November and December.

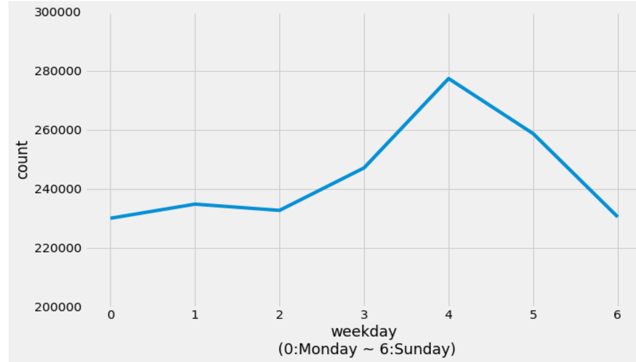

Fig 10. Weekday, group by whole data

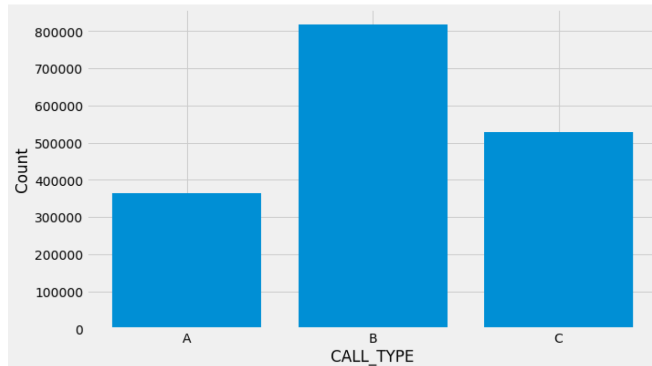

Fig 11. Call type

A. Trips dispatched from the central.

B. Trips that done directly to a taxi driver on a specific stand.

C. Otherwise (trip made on a random street).

In fig 3, May has the most number of trips whereas August has the least number of trips. In fig 4, January 1 and February 14 has the highest number of trips. In fig 5, March 1 and April 4 has the highest number of trips, whereas March 17 and April 21 has the least number of trips. In fig 6, May 9 and June 6 has the highest number of trips, whereas May 13 and June 10 recorded the least number of trips. In fig 7, July 5 and August 9 has the highest number of trips, whereas July 14 and August 17 saw least number of trips. In fig 8, September 26 and October 4 has the highest number of trips, least number of trips were on September 1 and October 20 and October 27. In fig 9, November 25 and December 20 has the highest number of trips, whereas November 17 to 18 and December 25 recorded the least number of trips. When we look into the graph which compares the trips according to the weekdays it can be said that Friday recorded the highest number of trips. In fig 10, graph shows that most number of trips were taken on Friday whereas Monday has the least number of trips. In fig 11, call type B has the most number of trips.

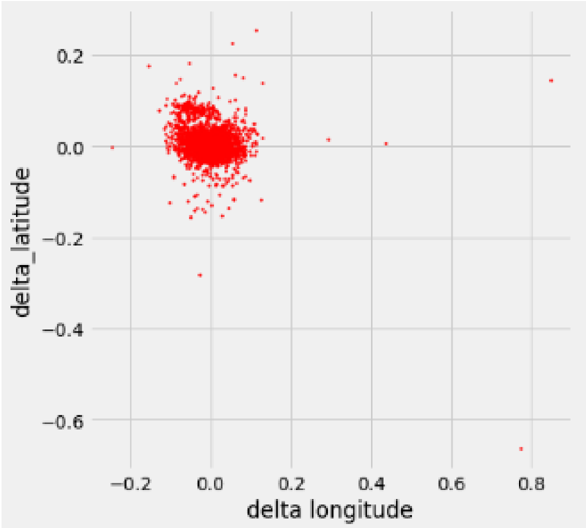

Fig 12. Distribution of delta latitude and longitude

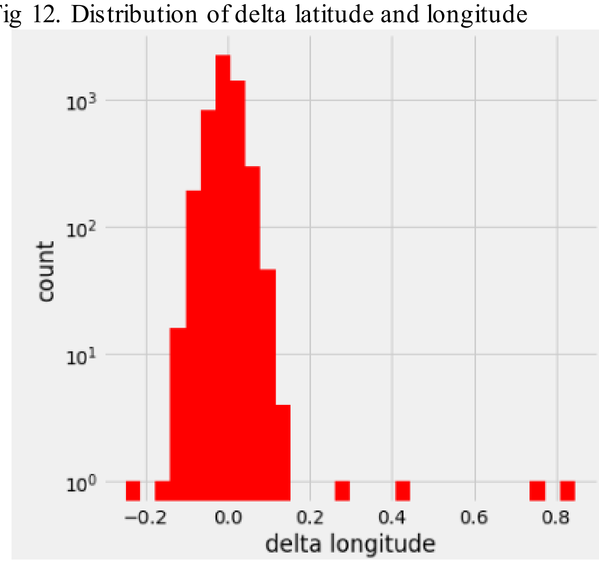

Fig 13. Distribution of Delta longitude

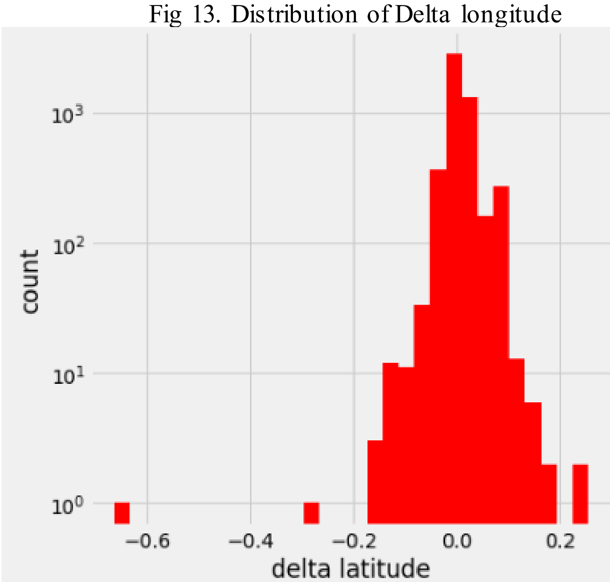

Fig 14. Distribution of Delta latitude

Distribution of travel distance and direction is done by finding Delta longitude and Delta latitude shown in Figures 12 -14. Delta longitude is calculated by subtracting lon_last with lon_1st. Delta latitude is calculated by subtracting lat_last with lat_1st.

Module 4: Regression analysis

Regression analysis is a set of statistical techniques used in data analysis to determine the connection between one or more independent variables and a dependent variable (also known as the outcome or response variable) (often called predictors, covariates, explanatory variables or features). The most common sort of regression analysis is linear regression, which involves finding the line (or a more sophisticated linear combination) that best fits the data based on a set of mathematical criteria. Based on values of independent variables, regression can predict the mean of dependent variables. Another technique to determine how two variables are related is to use correlation. It finds patterns in the dataset. The difference is that correlation evaluates the strength of an association between two variables, whereas regression characterizes the relationship's nature. Cluster analysis is performed to find similar patterns in the dataset. After analysis, topological structure is defined to analyze the dataset. Threshold and weights are calculated to predict the final destination.

To predict the point (longitude & latitude), We changed some features to categorical numerical values.

- Call type: A: 1, B: 2, C:3
- Origin_call: data exist: 1, null: 0
- Origin stand: data exist: 1, null: 0
- Day type: Only A, so this variable is no used.
- Missing data: True: 1, False=0

We're using Random Forest Regressor, Gradient Boosting Regressor, Linear Regression, K Neighbors Regressor, Decision Tree Regressor, Multi Output Regressor algorithms to predict the best route. Using grid search cv, we found that Linear regression is the best algorithm to be used with R2 score of 100%.

This method improves interpretability. Regularization is done to improve accuracy and generalization. It also has Automatic Variable Selection.

IV. CONCLUSION

In this study, we suggest the crossroads ranking approaches based on huge taxi trajectories are important. In contrast to prior research on the roles of nodes in road networks, this method makes use of dynamic network information. Then, to rank the significant crossroads, a score propagation algorithm over the tripartite network is proposed. The result indicated that, when matched to the standard approach, this algorithm is more likely to find significant crossroads with high connectivity, notably those located on circular roads, where a significant number of Source pairs originate and terminate.

V. DISCUSSION AND FUTURE WORK

The taxi searching and scheduling algorithms may be improved in the future by incorporating creditability of taxi drivers and riders This paper presents an approach by using the taxi driver's intelligence learned from an analysis of a large number of historic taxi routes data, we find the fastest route to a given departure with a given departure time. As an additional measure, we will reduce taxi travel distances further through ride sharing. In addition, we also propose introducing social constraints such as gender preference and habit preference, to refine the ride sharing mode.

VI. ACKNOWLEDGEMENT

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